

TrES-2b

R-1225

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190426 · created 2026-07-31T09:12:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458599.91692 +/- 0.00056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1195 +/- 0.0099
Transit depth (Rp/Rs)^2	1.43 +/- 0.24 [%]
Orbital Inclination (inc)	85.33 +/- 1.04
Airmass coefficient 1 (a1)	1.215 +/- 0.011
Airmass coefficient 2 (a2)	-0.1663 +/- 0.0075
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	5.48
Optimal Annulus	13.24
Transit Duration (day)	0.09 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1195 ± 0.0099 vs 0.1254 (deviation 0.31σ)
Duration fit vs published	0.09 d vs 0.0746 d (deviation 0.67σ)
Mid-time O–C	-1.6 min
Depth SNR	6.3
Residual scatter	0.86 %

Light curve

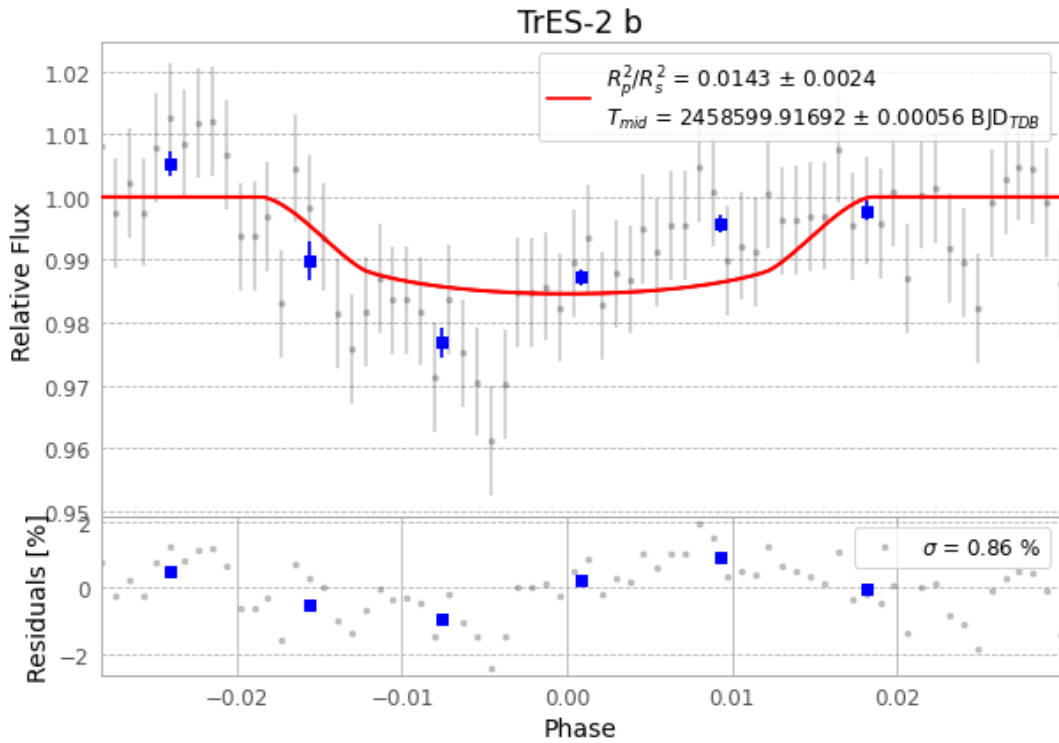


Figure 1 — FinalLightCurve_TrES-2 b_2019-04-26.png (SHA-256 ef9215cb8d8e392f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [440, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 72432178a2918d87...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2190426081812.FITS → TRES-2190426114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190426.log (54ef044b9bf1...), FinalLightCurve_TrES-2 b_2019-04-26.png (ef9215cb8d8e...), FinalLightCurve_TrES-2 b_2019-04-26.csv (d899b1a885f4...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)